# Docker Networking complete+Docker Swarm practicals

## 1. Custom Docker Networking with DNS-based Communication

```
Terminal

PS C:\Users\sreev> docker network ls
NETWORK ID     NAME                                      DRIVER    SCOPE
c1be007467d6   bridge                                    bridge    local
22b6e4fb751d   dnsdemo                                   bridge    local
e693e025f33d   dockercomposepractice1-nodemongo_default  bridge    local
f36f53361620   dockercomposepractice2_default            bridge    local
6a2d8cc7b1af   dockercompp3_default                      bridge    local
a19f70cacca7   dockercompp4_default                      bridge    local
98d3e2471d1f   forrahul_default                          bridge    local
cb273f47539e   host                                      host      local

RAM 2.40 GB  CPU 11.09%  Disk: 46.16 GB used (limit 1006.85 GB)
```

```
Terminal

PS C:\Users\sreev> docker network inspect bridge
[
    {
        "Name": "bridge",
        "Id": "c1be007467d66bd75f8355826c842877d3ec9f49e408bbcef634f4c53d45feec",
        "Created": "2026-04-17T10:00:01.818962208Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {

RAM 2.40 GB  CPU 5.92%  Disk: 46.16 GB used (limit 1006.85 GB)
```

```
Terminal

]
PS C:\Users\sreev> docker network create mynet
2516698b38b5000c6ea81cfe93383fc0d6d76be465ecf99590a2a18d3c312a8a
PS C:\Users\sreev> docker run -dit --name c1 --network mynet
docker: 'docker run' requires at least 1 argument

Usage:  docker run [OPTIONS] IMAGE [COMMAND] [ARG...]

See 'docker run --help' for more information
PS C:\Users\sreev> docker run -dit --name c1 --network mynet ubuntu
bfa0e2526bca231fbc8629828ec5ded3c26c073ccf2ca4182080a8328fc151a

RAM 2.40 GB  CPU 0.75%  Disk: 46.16 GB used (limit 1006.85 GB)
```

```
Terminal

PS C:\Users\sreev> docker run -dit --name c2 --network mynet ubuntu
3d0ad3f6030c1a88d7d47e349ed1a1f0e57a74cd259ddc4b1b3c854f7d0adedd
PS C:\Users\sreev> docker exec -it c1 bash
root@bfa0e2526bca:/# ping c
bash: ping: command not found
root@bfa0e2526bca:/# ping c2
bash: ping: command not found
root@bfa0e2526bca:/# apt update && apt install iputils-ping
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1507 kB]

RAM 2.40 GB  CPU 8.76%  Disk: 46.16 GB used (limit 1006.85 GB)
```

```
Terminal
debconf: falling back to frontend: Teletype
Setting up iputils-ping (3:20240117-1ubuntu0.1) ...
root@bfa0e2526bca:/# ping c2
PING c2 (172.27.0.3) 56(84) bytes of data.
64 bytes from c2.mynet (172.27.0.3): icmp_seq=1 ttl=64 time=0.997 ms
64 bytes from c2.mynet (172.27.0.3): icmp_seq=2 ttl=64 time=0.089 ms
64 bytes from c2.mynet (172.27.0.3): icmp_seq=3 ttl=64 time=0.109 ms
064 bytes from c2.mynet (172.27.0.3): icmp_seq=4 ttl=64 time=0.088 ms
64 bytes from c2.mynet (172.27.0.3): icmp_seq=5 ttl=64 time=0.232 ms
64 bytes from c2.mynet (172.27.0.3): icmp_seq=6 ttl=64 time=0.111 ms
64 bytes from c2.mynet (172.27.0.3): icmp_seq=7 ttl=64 time=0.115 ms
RAM 2.40 GB  CPU 0.83%  Disk: 46.16 GB used (limit 1006.85 GB)
>_ Terminal  Update available
```

## 2. Inter-Container Communication using Docker DNS (Backend–Database Connectivity)

```
Terminal
PS C:\Users\sreev> docker network create vishnu
c4cc63fa513ded6f9b1ae89a68579ad083a69d9e62f12c54f4402595ba8d2673
PS C:\Users\sreev> docker run -d --name mysqlcont --network vishnu -e MYSQL_ROOT_PASSWORD=root -e MYSQL_DATABASE=testdb mysql:8
e13f70ae7de5bfb190e7801ad17cb42b5a7272fe942977cd5b02dde2dc70283d
PS C:\Users\sreev> docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED          STATUS                          PORTS                   NAMES
e13f70ae7de5   mysql:8               "docker-entrypoint.s…"   22 seconds ago   Up 21 seconds                   3306/tcp, 33060/tcp     mysqlcont
035757ee2d48   0236ee02dcbc          "/docker-entrypoint.…"   3 weeks ago      Restarting (1) 16 seconds ago                           nginx_gateway
36532afd82dc   dockercompp4-backend  "python app.py"          3 weeks ago      Up About an hour                                        python_backend
ac9a06a43887   0236ee02dcbc          "/docker-entrypoint.…"   3 weeks ago      Restarting (1) 15 seconds ago                           gateway
PS C:\Users\sreev> docker run -dit --name backend --network vishnu ubuntu
c63d31b2bb1a3cfc6ae25235ae3a549301f61fd1a25e46fc0b6c79dcfa4d1421
PS C:\Users\sreev> docker exec -it backend bash
RAM 2.76 GB  CPU 0.25%  Disk: 46.48 GB used (limit 1006.85 GB)
>_ Terminal  Update available
```

```
Terminal
PS C:\Users\sreev> docker run -dit --name backend --network vishnu ubuntu
c63d31b2bb1a3cfc6ae25235ae3a549301f61fd1a25e46fc0b6c79dcfa4d1421
PS C:\Users\sreev> docker exec -it backend bash
root@c63d31b2bb1a:/# apt update
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [3581 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1808 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [34.8 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [2025 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
RAM 2.77 GB  CPU 6.38%  Disk: 46.48 GB used (limit 1006.85 GB)
>_ Terminal  Update available
```

```
Terminal
root@c63d31b2bb1a:/# apt install -y mysql-client
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libbsd0 libedit2 mysql-client-8.0 mysql-client-core-8.0 mysql-common
The following NEW packages will be installed:
  libbsd0 libedit2 mysql-client mysql-client-8.0 mysql-client-core-8.0 mysql-common
0 upgraded, 6 newly installed, 0 to remove and 15 not upgraded.
Need to get 2918 kB of archives.
After this operation, 62.2 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libbsd0 amd64 0.12.1-1build1.1 [41.2 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libedit2 amd64 3.1-20230828-1build1 [97.6 kB]
RAM 2.76 GB  CPU 6.95%  Disk: 46.48 GB used (limit 1006.85 GB)
>_ Terminal  Update available
```

```
Terminal
Preparing to unpack .../5-mysql-client_8.0.45-0ubuntu0.24.04.1_all.deb ...
Unpacking mysql-client (8.0.45-0ubuntu0.24.04.1) ...
Setting up mysql-common (5.8+1.1.0build1) ...
update-alternatives: using /etc/mysql/my.cnf.fallback to provide /etc/mysql/my.cnf (my.cnf) in auto mode
Setting up libbsd0:amd64 (0.12.1-1build1.1) ...
Setting up libedit2:amd64 (3.1-20230828-1build1) ...
Setting up mysql-client-core-8.0 (8.0.45-0ubuntu0.24.04.1) ...
Setting up mysql-client-8.0 (8.0.45-0ubuntu0.24.04.1) ...
Setting up mysql-client (8.0.45-0ubuntu0.24.04.1) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
root@c63d31b2bb1a:/# mysql -h localhost -u root -p
Enter password:
ERROR 2002 (HY000): Can't connect to local MySQL server through socket '/var/run/mysqld/mysqld.sock' (2)
RAM 2.76 GB  CPU 6.26%  Disk: 46.48 GB used (limit 1006.85 GB)
>_ Terminal  Update available
```

- **Some context about the above pic, if you try to use localhost to connect, it does not refer to your host machine or other containers. Inside a Docker container, localhost represents the container itself. That's why using localhost to connect to services like MySQL running in another container fails. Instead, we should use the container name (like MySQL), which Docker resolves using its internal DNS.**

```
Terminal
root@c63d31b2bb1a:/# mysql -h mysqlcont -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.4.8 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
RAM 2.76 GB  CPU 0.42%  Disk: 46.48 GB used (limit 1006.85 GB)
```
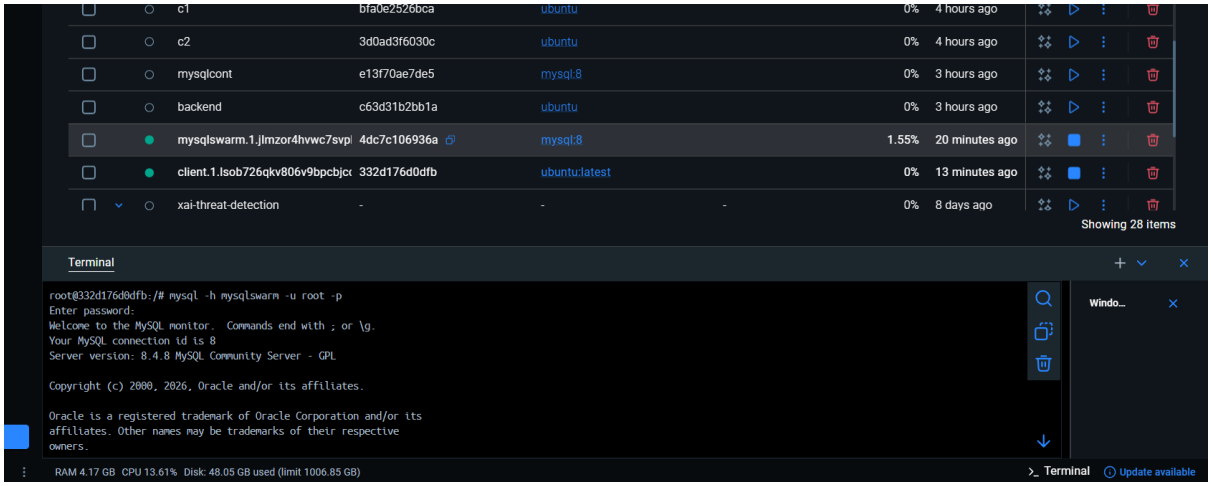

```
Terminal
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
| testdb             |
+--------------------+
5 rows in set (0.08 sec)
RAM 2.76 GB  CPU 5.44%  Disk: 46.48 GB used (limit 1006.85 GB)
```

## 3. Implementation of Docker Swarm with Overlay Network for Service-to-Service Communication

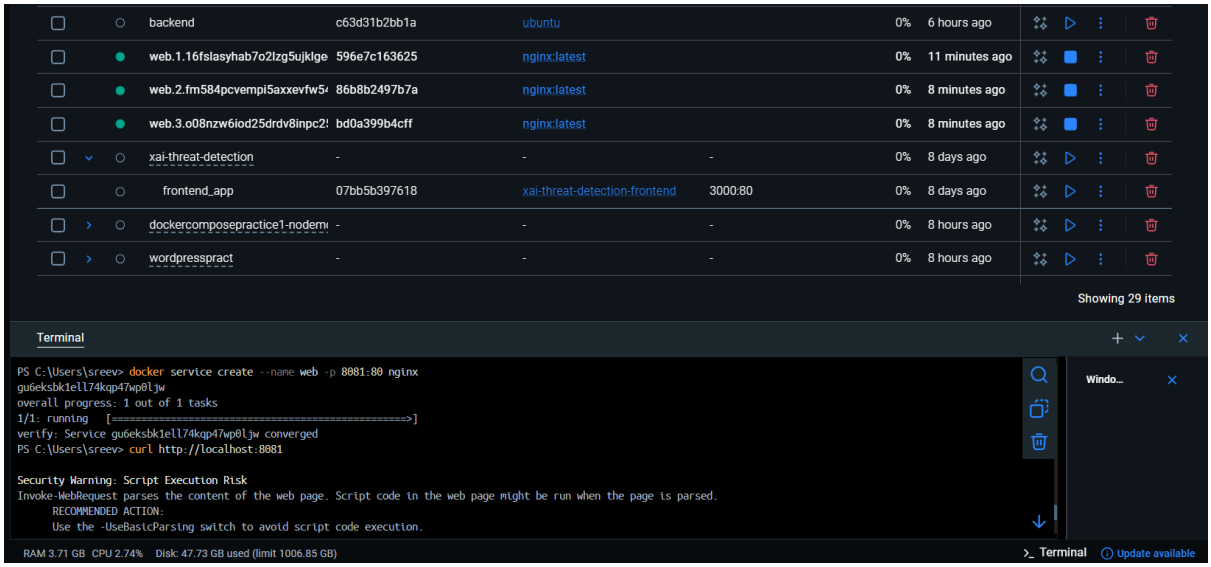
```
Terminal
PS C:\Users\sreev> docker swarm init
Swarm initialized: current node (e53zt4afbqr9dhtnlkzvdvt6k) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-4fu97i1grl75unyhjac70ugll7xq4xzmjspmjkptktvjnk5pwo-8fk18ahczltpk4cdkg9r2f4a5 192.168.65.3:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
PS C:\Users\sreev> docker network create -d overlay myoverlay
oucv9qmaxxk2m8p929b9gfgdk
PS C:\Users\sreev> docker service create --name mysqlswarm --network myoverlay -e MYSQL_ROOT_PASSWORD=root mysql:8
RAM 4.15 GB  CPU 15.84%  Disk: 48.05 GB used (limit 1006.85 GB)
```

```
Terminal
gmg3pxv06yss6u55zp5asats4
overall progress: 1 out of 1 tasks
1/1: running   [==================================================>]
verify: Service gmg3pxv06yss6u55zp5asats4 converged
PS C:\Users\sreev> docker service create --name client --network myoverlay ubuntu sleep infinity
vij4o2bctrs2desv0tqtfdo71
overall progress: 1 out of 1 tasks
1/1: running   [==================================================>]
verify: Service vij4o2bctrs2desv0tqtfdo71 converged
PS C:\Users\sreev> docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED         STATUS                       PORTS                 NAMES
RAM 4.16 GB  CPU 0.33%  Disk: 48.05 GB used (limit 1006.85 GB)
```

```
Terminal
CONTAINER ID   IMAGE                 COMMAND                  CREATED         STATUS                       PORTS                 NAMES
332d176d0dfb   ubuntu:latest         "sleep infinity"         6 minutes ago   Up 6 minutes                                       client.1.lsob726qkv806v9bpcbjcc3qe
4dc7c106936a   mysql:8               "docker-entrypoint.s…"   13 minutes ago  Up 13 minutes                3306/tcp, 33060/tcp   mysqlswarm.1.jlmzor4hvwc7svphapnmql
o4d
035757ee2d48   0236ee02dcbc          "/docker-entrypoint.…"   3 weeks ago     Restarting (1) 48 seconds ago                      nginx_gateway
36532afd82dc   dockercompp4-backend  "python app.py"          3 weeks ago     Up 4 hours                                         python_backend
ac9a06a43887   0236ee02dcbc          "/docker-entrypoint.…"   3 weeks ago     Restarting (1) 47 seconds ago                      gateway
PS C:\Users\sreev> docker exec -it 332d176d0dfb bash
root@332d176d0dfb:/# apt update && apt install -y mysql-client
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
RAM 4.17 GB  CPU 14.04%  Disk: 48.05 GB used (limit 1006.85 GB)
```




# 4. Implementation of Service Scaling and Load Balancing in Docker Swarm using Routing Mesh



```
Terminal
PS C:\Users\sreev> docker exec -it 86b8b2497b7a bash
root@86b8b2497b7a:/# echo "Hello from container 1">/usr/share/nginx/html/index.html
root@86b8b2497b7a:/# exit;
exit
PS C:\Users\sreev> docker exec -it bd0a399b4cff bash
root@bd0a399b4cff:/# echo "Hello from container 2">/usr/share/nginx/html/index.html
root@bd0a399b4cff:/# exit;
exit
PS C:\Users\sreev> docker exec -it 596e7c163625 bash
root@596e7c163625:/# echo "Hello from container 3">/usr/share/nginx/html/index.html
root@596e7c163625:/# exit;
RAM 3.70 GB  CPU 1.67%  Disk: 47.73 GB used (limit 1006.85 GB)
Terminal  Update available
```

```
Terminal
PS C:\Users\sreev> docker exec -it 596e7c163625 bash
root@596e7c163625:/# echo "Hello from container 3">/usr/share/nginx/html/index.html
root@596e7c163625:/# exit;
exit
PS C:\Users\sreev> curl.exe http://localhost:8081
Hello from container 1
PS C:\Users\sreev> curl.exe http://localhost:8081
Hello from container 2
PS C:\Users\sreev> curl.exe http://localhost:8081
Hello from container 3
PS C:\Users\sreev>
RAM 3.70 GB  CPU 0.08%  Disk: 47.73 GB used (limit 1006.85 GB)
Terminal  Update available
```

**Some notes for this:**

- Global Port Accessibility: When you publish a service port (e.g., port 8081), Docker binds that port on every node in the swarm. An external request hitting any node's IP on that port is automatically routed to a healthy container running on any node in the cluster.
- Automatic Load Balancing: It uses the Linux kernel's IPVS (IP Virtual Server) to distribute incoming traffic across all service replicas in a round-robin fashion.
- Service Discovery: It integrates with Docker's internal DNS, allowing services within the swarm to reach each other using their service names rather than static IP addresses.